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48. Suppose that the uncertainty in determining the position of an electron in an orbit is 0.6 Å. What is the uncertainty in its momentum?

49. Show that if the measurement of the uncertainty in the location of the particle is equal to its de Broglie wavelength, the minimum uncertainty in its velocity is equal to its velocity / $4\pi$

50. What is the de Broglie wave length of an electron, which is accelerated from the rest, through a potential difference of 100V?

51. Identify the missing quantum numbers and the sub energy level

| n | l | m | Sub energy level |
|---|---|---|---|
| ? | ? | 0 | 4d |
| 3 | 1 | 0 | ? |
| ? | ? | ? | 5p |
| ? | ? | -2 | 3d |